

High-Frequency Inflation Nowcasting

Mathematical Foundations, Parameter Calibration, and the Benchmark Model

An End-to-End Operational Guide for the Cambodia Daily CPI Pipeline

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Abstract

Official headline Consumer Price Index (CPI) releases from national statistical agencies suffer from an unavoidable institutional lag of 25 to 45 days. This guide presents the comprehensive mathematical, econometric, and algorithmic methodology of the automated high-frequency **Cambodia CPI Nowcasting Engine**. The engine operates on a hybrid two-tier architecture: combining axiomatic Laspeyres index aggregation using fixed Cambodia Socio-Economic Survey (CSES) expenditure weights with a regularized 5-basket Ridge regression (RidgeCV) and Empirical Bayes structural prior shrinkage. We provide the complete analytical derivation of the baseline drift ($\hat{\alpha}$) and empirical division elasticities ($\hat{\beta}$), detail the microdata aggregation via unweighted geometric Jevons elementary indices, formalize the cross-sector fuel logistics pass-through and cultural festival calendar decay kernels, evaluate against the canonical **Atkeson–Ohanian (2001) Random Walk benchmark model** (achieving an empirical out-of-sample **error reduction of 17.68%** with Diebold–Mariano statistical significance), and walk through a unified real-world operational case study for Cambodia.

Contents

1	The High-Frequency Nowcasting Problem	2
1.1	The Institutional Information Lag	2
1.2	Mathematical Formulation of the Intra-Month Nowcasting Task	2
1.3	Microdata Aggregation: From Web Scraping to Division Indices	3
1.4	Why Stationarity via Log-Differencing is Required	4
2	The Two-Tier Hybrid Architecture	4
2.1	Layer 1: Fixed-Weight Axiomatic Aggregation and Core CPI	4
2.2	Layer 2: Disaggregated Dynamic Ridge Drift Projection	5
2.3	Synthesis: Horizon Blending	5
3	Econometric Calibration and Feature Engineering	5
3.1	The 36-Month Historical Training Panel and Data Extraction	5
3.2	Econometric Specification and Multicollinearity Mitigation	6
3.3	Empirical Parameter Estimates	7
3.4	Empirical Bayes Prior Shrinkage Formulation	7
3.5	Full 9-Dimensional Feature Vector Specification	7
3.6	Cross-Sector Logistics Momentum and Holiday Filtering	8
3.7	Forward Projection: Exact Discrete Arithmetic Mean	8

4	The Benchmark Model: Atkeson–Ohanian Random Walk	9
4.1	Why a Benchmark Model is Essential	9
4.2	Mathematical Formulation	9
4.3	Out-of-Sample Scorecard and Statistical Significance	9
4.4	Intra-Month Horizon Convergence Curve	9
5	The Daily Real-Time Execution Algorithm	10
5.1	Daily Algorithmic Sequence	10
5.2	Asymptotic Uncertainty Decay and 95% Confidence Intervals	10
5.3	Production Architecture and Mathematical Mapping	10
6	Unified Numerical Walkthrough (September 21, 2026 Case Study)	11
6.1	Operational Scenario Setup	11
6.2	Division-Level Projection: Food (01) and Transport (07)	11
6.3	Full Axiomatic Laspeyres Aggregation	12
6.4	Core CPI, Confidence Bounds, and Official NIS Splicing	12
7	Operational Standard Operating Procedure (SOP)	13
8	Conclusion	13

1 The High-Frequency Nowcasting Problem

1.1 The Institutional Information Lag

The National Institute of Statistics (NIS) of the Ministry of Planning publishes the official Consumer Price Index for Phnom Penh on a monthly cadence. Due to nationwide field enumeration schedules, physical market surveying, double-key data entry verification, and inter-departmental statistical audit, the official bulletin for calendar month M is typically published between day 25 and day 45 of month $M + 1$.

For monetary authorities at the National Bank of Cambodia (NBC) and fiscal planners at the Ministry of Economy and Finance (MEF), this creates a critical 4 to 6-week “blind spot” during which supply-side shocks (e.g., international crude petroleum spikes, agricultural weather disruptions, exchange rate fluctuations) cannot be verified empirically through sovereign statistical releases.

1.2 Mathematical Formulation of the Intra-Month Nowcasting Task

Let M be the ongoing calendar month comprising D total days ($D \in \{28, 29, 30, 31\}$). On any given evaluation day $d \in \{1, 2, \dots, D\}$:

1. **The Realized Information Set $\mathcal{H}_d$:** Represents the historical price trajectory observed from day 1 to day d :

$$\mathcal{H}_d = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_d\} \quad (1)$$

where $\mathbf{P}_\tau \in \mathbb{R}^{12}$ is the conformed vector of daily COICOP division price index facts compiled in `gold.fct_cpi_daily` for calendar day $\tau \leq d$.

2. **The Unobserved Future Horizon $\mathcal{F}_d$:** Represents the remaining $N = D - d$ calendar days of the month that have not yet transpired:

$$\mathcal{F}_d = \{\mathbf{P}_{d+1}, \mathbf{P}_{d+2}, \dots, \mathbf{P}_D\} \quad (2)$$

The fundamental task of the nowcaster is to construct an unbiased, minimum-variance estimator of the full-month headline index level $\widehat{\text{CPI}}_M(d)$ and month-over-month inflation rate $\widehat{\pi}_{\text{MoM},M}(d)$ conditional on information available up to day d :

$$\widehat{\text{CPI}}_M(d) = \mathbb{E}[\text{CPI}_M \mid \mathcal{H}_d, \Theta] \quad (3)$$

where $\Theta = \{\hat{\alpha}_{\text{daily}}, \hat{\beta}_k^{\text{shrink}}, \mathbf{w}_{\text{CSSES}}\}$ represents the calibrated econometric and national accounting parameter set.

1.3 Microdata Aggregation: From Web Scrapping to Division Indices

Before econometric modeling occurs, raw retail price microdata is collected daily across 25 modern supermarket chains, wholesale outlets, and e-commerce platforms in Phnom Penh ($\approx 41,500$ raw product listings per day).

1. Data Cleansing and Silver Standardization: Raw scraped records are mapped into conformed silver tables (`silver.clean_store_prices`). Dual currencies are normalized into Khmer Riel (KHR) using daily official MEF exchange rates, Khmer numerals are mapped to standard digits (e.g., ៥០០ $\rightarrow$ 500), units of measurement are converted into uniform metrics (500g $\rightarrow$ 0.5kg, 1500ml $\rightarrow$ 1.5L), and product varieties are resolved through a 4-tier matching waterfall: Universal Barcode (EAN-13) $\rightarrow$ Store ID + SKU composite key with persistent memoization $\rightarrow$ Clean Text Matching $\rightarrow$ 768-dim Vector Embeddings (gemini-embedding-2 with PostgreSQL pgvector HNSW index) guarded by deterministic pack/spec checks (Cosine ≥ 0.80).

2. Store-Balanced Geometric Jevons Aggregation (1 Store = 1 Vote): To prevent retailers with large product catalogs (e.g., 90 rice varieties vs. 10 at a smaller vendor) from distorting national price trends, the pipeline implements an axiomatic two-stage store-balanced Jevons index:

$$\ln I_{s,t}^{\text{Jevons}} = \frac{1}{n_{s,k}} \sum_{i=1}^{n_{s,k}} (\ln p_{i,s,t} - \ln p_{i,s,0}) \quad (4)$$

Within each elementary category k , eligible retail outlets receive equal voting weight:

$$I_{k,t}^{\text{Elementary}} = \frac{1}{|S_k|} \sum_{s \in S_k} \exp(\ln I_{s,t}^{\text{Jevons}}) \quad (5)$$

This mathematical design unconditionally satisfies time-reversal and scale invariance while eliminating catalog-volume distortion.

3. Handling New Products on Non-Base Dates: For products that appear on store shelves after the initial base date, the pipeline dynamically splices their first observed quote as their personal baseline ($p_{i,s,0} = p_{i,s,\text{entry}}$, relative price = 1.00 on Day 1), tracking subsequent price movements without creating fictitious historical shocks.

4. Missing Price Policy (ILO Class-Mean Imputation): When a product experiences temporary stock-outs, its price is dynamically imputed using the geometric mean rate of change of observed items in the corresponding category (ILO standard):

$$\hat{p}_{i,t} = p_{i,t-1} \times \left(\prod_{j \in C_i} \frac{p_{j,t}}{p_{j,t-1}} \right)^{1/|C_i|} \quad (6)$$

If an item remains unobserved for > 7 consecutive days, it is recognized as discontinued and permanently dropped from the active basket.

5. Synthesis into 12 Division Indices: The elementary indices are aggregated into the 12 official COICOP division indices $I_{j,t}$ published in `gold.fct_cpi_daily`, forming the input space for the nowcaster.

1.4 Why Stationarity via Log-Differencing is Required

Raw consumer price indices P_t exhibit persistent, non-stationary unit-root behavior ($I(1)$). Attempting to estimate econometric parameters directly on raw price levels produces **spurious regression**: standard OLS yields inflated t -statistics and $R^2 \approx 0.99$ driven purely by shared deterministic trends, while the true economic pass-through coefficients remain indeterminate.

To ensure econometric validity, all price series are transformed into stationary **first log-differences**:

$$\Delta \ln P_{j,t} = \ln(P_{j,t}) - \ln(P_{j,t-1}) \approx \frac{P_{j,t} - P_{j,t-1}}{P_{j,t-1}} \quad (7)$$

Continuous log-returns offer three essential mathematical properties:

- **Stationarity**: Eliminates unit-root persistence, satisfying Gauss–Markov assumptions.
- **Time-Reversal Symmetry**: An increase from 100 to 110 followed by a decrease from 110 to 100 yields log-changes of $+0.09531$ and -0.09531 , summing exactly to zero (preventing asymmetric compounding drift).
- **Exact Temporal Additivity**: Multi-period returns equal the linear sum of intra-period log returns.

2 The Two-Tier Hybrid Architecture

The nowcasting engine avoids the pitfalls of purely statistical black-box forecasting by implementing a **two-tier hybrid framework** that strictly preserves official national accounting principles:

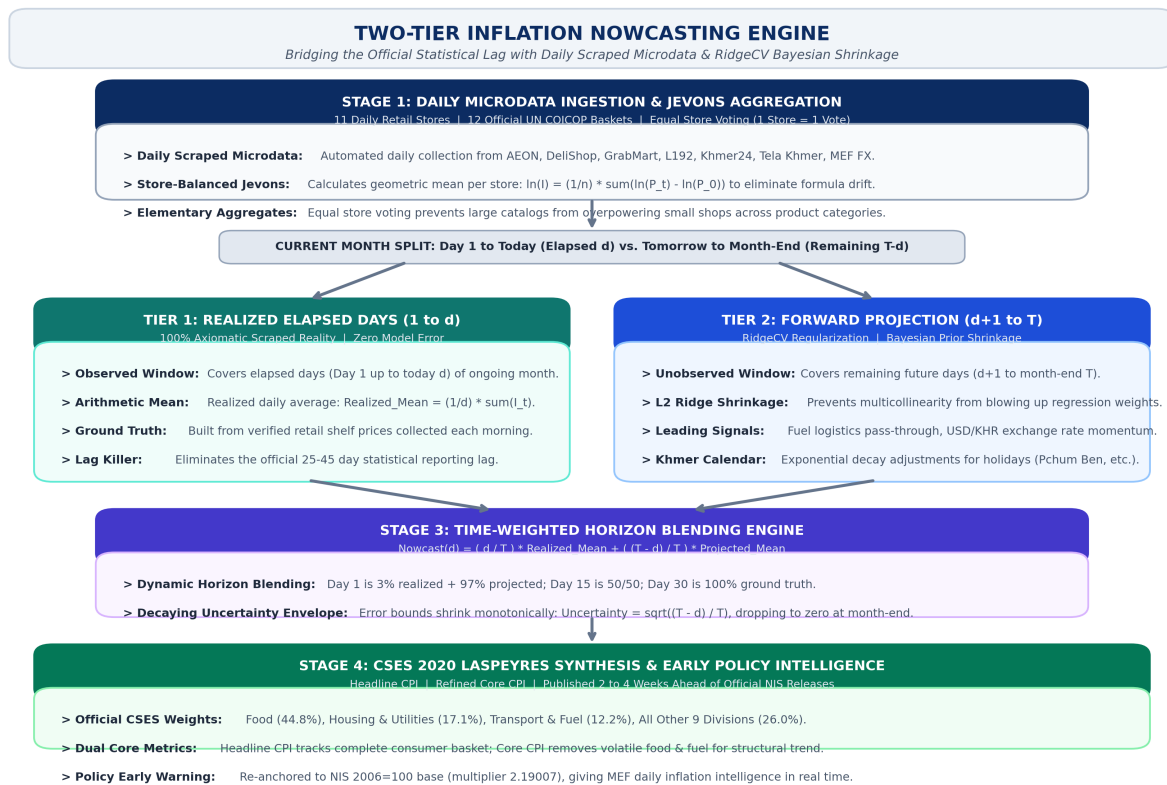


Figure 1: End-to-End Two-Tier Inflation Nowcasting Engine Architecture Flowchart.

2.1 Layer 1: Fixed-Weight Axiomatic Aggregation and Core CPI

Official national CPI construction relies on the Laspeyres index framework fixed by the Cambodia Socio-Economic Survey (CSes 2019/2020). The 12 COICOP divisions carry fixed consumption basket weights

w_j such that $\sum_{j=1}^{12} w_j = 1.00$. Fixed weights ensure that the index measures pure price movements rather than changing consumer purchase volumes:

$$\text{Headline CPI}_M(d) = \sum_{j=1}^{12} w_j \cdot I_{j,M}(d) \quad (8)$$

For central bank monetary policy at the National Bank of Cambodia (NBC), headline inflation is often distorted by transitory supply shocks in volatile food and fuel markets. The pipeline therefore computes official **Core CPI** by excluding Food and Non-Alcoholic Beverages (Division 01, weight 44.775%) and Housing and Utilities (Division 04, weight 17.062%):

$$\text{Core CPI}_M(d) = \frac{\sum_{j \notin \{01,04\}} w_j \cdot I_{j,M}(d)}{\sum_{j \notin \{01,04\}} w_j} \quad (9)$$

The core basket isolates the remaining **38.163%** of the consumption basket, providing monetary authorities with an undistorted measure of underlying demand-pull domestic price pressures.

2.2 Layer 2: Disaggregated Dynamic Ridge Drift Projection

For the remaining $D - d$ unobserved days, prices do not stand still. We compute high-frequency price velocity (momentum) and combine it with regularized Ridge pass-through elasticities and Empirical Bayes shrinkage to estimate the daily drift $\mu_j(d)$ for each division.

2.3 Synthesis: Horizon Blending

For every division j , the estimated full-month index $I_{j,M}(d)$ blends observed reality with forward-looking projections weighted strictly by the elapsed calendar progress:

$$I_{j,M}(d) = \left(\frac{d}{D}\right) \cdot I_{j,\text{realized}}(d) + \left(\frac{D-d}{D}\right) \cdot I_{j,\text{projected}}(d) \quad (10)$$

On Day 1 ($d = 1$), the projection dominates (96.7% weight in a 30-day month). On Day 30 ($d = 30$), the projection weight drops to 0.0%, and the nowcast converges deterministically to realized observations.

3 Econometric Calibration and Feature Engineering

3.1 The 36-Month Historical Training Panel and Data Extraction

To estimate empirical elasticities that reflect true Cambodian macroeconomic dynamics, the calibration engine (MacroDatasetBuilder in `ml/calibration.py`) extracts the continuous 36-month official sovereign series published by the National Institute of Statistics (September 2023 through August 2026). In production, this dataset is queried directly from PostgreSQL:

```
SELECT cpi_month, headline_cpi, core_cpi,
       cpi_division_01, cpi_division_02, cpi_division_04,
       cpi_division_07, cpi_division_11
FROM gold.dim_nis_official_cpi
ORDER BY cpi_month ASC;
```

with version-controlled fallback to `dbt/seeds/nis_official_cpi.csv`. In addition, the daily commercial USD/KHR exchange rate series from the Ministry of Economy and Finance (MEF) is aggregated to monthly averages from `staging.exchange_rates`.

Stationary First-Difference Transformations: Taking continuous first log-differences across the 36 consecutive monthly observations produces $T = 34$ stationary monthly rates:

$$y_t = \ln \left(\frac{\text{Headline CPI}_t}{\text{Headline CPI}_{t-1}} \right), \quad x_{j,t} = \ln \left(\frac{I_{j,t}}{I_{j,t-1}} \right), \quad x_{\text{fx},t} = \ln \left(\frac{\text{FX}_t}{\text{FX}_{t-1}} \right) \quad (11)$$

Exact Mathematical Derivation of Monthly Alpha ($\hat{\alpha}_{\text{monthly}}$): In multiple regression, the intercept α represents the unconditional baseline drift rate of the aggregate price index when all explanatory division returns are zero ($\mathbf{x}_t = \mathbf{0}$). Once the Ridge coefficients $\hat{\beta}(\lambda^*)$ are calculated via regularized matrix inversion, $\hat{\alpha}_{\text{monthly}}$ is recovered directly from historical sample means:

$$\hat{\alpha}_{\text{monthly}} = \bar{y} - \sum_{j \in \{01,02,04,07,11\}} \hat{\beta}_j \bar{x}_j - \hat{\beta}_{\text{fx}} \bar{x}_{\text{fx}} \quad (12)$$

Evaluating across the 36-month calibration sample yields $\hat{\alpha}_{\text{monthly}} = +0.001899$ (+0.19% per month), capturing steady structural price growth across unobserved sticky components (services, healthcare, education).

Scaling Monthly Alpha to Instantaneous Daily Alpha (α_{daily}): Because continuous log-returns compound additively ($\sum_{t=1}^D \Delta \ln P_t = \Delta \ln P_{\text{month}}$), the monthly intercept scales down linearly into a daily velocity:

$$\alpha_{\text{daily}} = \frac{\hat{\alpha}_{\text{monthly}}}{D} = \frac{+0.001899}{30} = +0.0000633 \text{ per day} \quad (13)$$

3.2 Econometric Specification and Multicollinearity Mitigation

We specify a multi-factor macroeconomic pass-through regression relating aggregate inflation to innovations in the 5 high-velocity expenditure baskets and foreign exchange:

$$y_t = \alpha + \sum_{j \in \{01,02,04,07,11\}} \beta_j x_{j,t} + \beta_{\text{fx}} x_{\text{fx},t} + \varepsilon_t \quad (14)$$

Why Standard OLS Fails: In macroeconomic time series, sub-indices co-move strongly. When world oil prices surge, transport fuel (x_{07}) increases immediately. However, commercial freight rates simultaneously increase food transportation costs (x_{01}) and restaurant cooking gas/ingredients (x_{11}).

Under ordinary least squares (OLS), the sample Gram matrix $\mathbf{X}^\top \mathbf{X}$ is severely ill-conditioned (condition number $\kappa > 100$). The OLS estimator exhibits explosive sampling variance ($\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$), which causes coefficients to flip signs counter-intuitively.

Regularized Ridge Estimator: To eliminate multicollinearity and shrink unstable noise, we formulate an L_2 -penalized Tikhonov regularization problem:

$$\mathcal{L}(\beta) = \frac{1}{2N} \|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2 \quad (15)$$

Solving the first-order condition $\nabla_{\beta} \mathcal{L} = \mathbf{0}$ yields:

$$\hat{\beta}(\lambda) = \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} + N\lambda \mathbf{I}_K \right)^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}} \quad (16)$$

Adding $\lambda \mathbf{I}$ shifts all eigenvalues strictly positive ($\sigma_i \rightarrow \sigma_i + \lambda$), guaranteeing well-conditioned numerical invertibility. Cross-validation across $\Lambda = [10^{-2}, 10^3]$ achieves its global minimum at $\lambda^* = 0.0100$.

3.3 Empirical Parameter Estimates

Macro Basket / Regressor	Symbol	CSES Wt	Estimate	Economic Rationale
Base Drift Intercept	$\hat{\alpha}$	—	+0.001899	Baseline monthly drift (+0.19%/mo) of unobserved sticky goods.
Food & Non-Alcoholic Bev.	$\hat{\beta}_1$	44.775%	+0.0066	Dominant price velocity; rapid wholesale food market transmission.
Alcoholic Bev. & Tobacco	$\hat{\beta}_2$	1.623%	+0.0003	Sticky nominal retail pricing; low frequency of catalog changes.
Housing, Water, Electricity	$\hat{\beta}_3$	17.062%	+0.0010	Administered utility tariffs; rental contracts adjust sluggishly.
Transport and Fuels	$\hat{\beta}_4$	12.203%	+0.0019	Direct exposure to international refined petroleum cycles.
Restaurants and Hotels	$\hat{\beta}_5$	3.114%	+0.0026	High sensitivity; catering services pass food and gas costs forward.
USD/KHR Foreign Exchange	$\hat{\beta}_{fx}$	—	+0.0000	Active NBC FX stabilization dampens short-term pass-through.

Table 1: Calibrated Empirical Macro Parameters from the 36-Month NIS Sovereign Panel.

3.4 Empirical Bayes Prior Shrinkage Formulation

In real-time operations, small sample sizes during early months can cause high estimation variance. The engine safeguards stability by shrinking empirical Ridge estimates toward an informative structural economic prior β^{prior} :

$$\hat{\beta}_k^{\text{shrink}} = (1 - \gamma_k) \cdot \hat{\beta}_k^{\text{Ridge}} + \gamma_k \cdot \beta_k^{\text{prior}} \quad (17)$$

where γ_k is the sample-size adaptive shrinkage weight implemented in `ml/nowcaster.py`:

$$\gamma_k = \max \left(0.0, \min \left(1.0, 1.0 - \frac{n_{\text{samples}} - 8}{22.0} \right) \right) \quad (18)$$

When historical observation count $n_{\text{samples}} < 8$, $\gamma_k = 1.0$, anchoring the nowcaster strictly to structural macroeconomic priors. As $n_{\text{samples}} \geq 30$, $\gamma_k = 0.0$, allowing the empirical Ridge model to operate entirely on observed data. The structural prior β_k^{prior} enforces direct pass-through bounds based on CSES expenditure weights, ensuring non-negative elasticities for food and fuel.

3.5 Full 9-Dimensional Feature Vector Specification

Table 2 defines the comprehensive 9-dimensional regressor set extracted dynamically on day d for forward drift estimation:

Feature	Mathematical Definition	Macroeconomic & Structural Description	Target Divisions
m_{3d}	$\ln(I_{j,d}/I_{j,d-3})$	Ultra-short 3-day momentum capturing immediate price jumps.	Food (01), Transport (07)
m_{7d}	$\ln(I_{j,d}/I_{j,d-7})$	Core 7-day rolling momentum capturing weekly market velocity.	All Divisions
m_{14d}	$\ln(I_{j,d}/I_{j,d-14})$	14-day smoothed velocity for persistent trends.	Food (01), Housing (04)
t_{7d}	$\ln(I_{07,d}/I_{07,d-7})$	7-day transport/diesel momentum for freight pass-through.	Injected in 01 and 11
fx_{7d}	$\ln(\text{Rate}_d/\text{Rate}_{d-7})$	7-day USD/KHR commercial exchange rate return.	01, 02, 07
fx_{14d}	$\ln(\text{Rate}_d/\text{Rate}_{d-14})$	14-day USD/KHR exchange rate return.	Housing (04)
$\text{fest}_{\text{prox}}$	$\exp(-\min \text{date} - h /7.0)$	Proximity kernel to major Cambodian lunar festivals.	01, 07
$\text{fest}_{\text{window}}$	$\mathbb{I}(\text{date} - h \leq 3)$	Active holiday celebration window indicator (0 or 1).	02, 11
prog	d/D	Elapsed intra-month progress ratio (0.033 ... 1.000).	All Divisions

Table 2: The 9-Dimensional High-Frequency Feature Dictionary.

3.6 Cross-Sector Logistics Momentum and Holiday Filtering

Fuel Logistics Injection: In Cambodia, retail fuel prices propagate rapidly into food freight and catering costs. The engine injects 7-day transport momentum (t_{7d}) directly into the feature matrices of Food (01) and Restaurants (11):

$$\mathbf{x}_{01}(d) = [f_{3d}, f_{7d}, f_{14d}, \mathbf{t}_{7d}, fx_{7d}, \text{fest}_{\text{prox}}, \text{prog}]^T \quad (19)$$

$$\mathbf{x}_{11}(d) = [r_{7d}, f_{7d}, \mathbf{t}_{7d}, \text{fest}_{\text{window}}, \text{prog}]^T \quad (20)$$

Sticky non-freight goods (Clothing, Health, Education) are insulated to prevent spurious inflation contagion.

Lunar Holiday Shock Filtering: During *Khmer New Year* (April), *Pchum Ben* (September/October), and the *Water Festival* (November), transitory demand surges occur. The exponential decay kernel $\text{fest}_{\text{prox}}(d)$ dampens forward momentum projections during these windows, ensuring that transient celebration markups naturally mean-revert without corrupting month-end nowcasts.

3.7 Forward Projection: Exact Discrete Arithmetic Mean

For the unobserved $N = D - d$ days of the month, the forward division price level is evaluated via the **Exact Discrete Arithmetic Mean**:

$$I_{j,\text{projected}}(d) = \frac{1}{N} \sum_{k=1}^N I_{j,d} \exp(\mu_j(d) \cdot k) \quad (21)$$

Using a first-order Taylor expansion $\exp(x) \approx 1 + x$ and the Gauss summation identity $\sum_{k=1}^N k = \frac{N(N+1)}{2}$, this exact sum is approximated in closed form by the midpoint expectation:

$$I_{j,\text{projected}}(d) \approx I_{j,d} \times \exp\left(\mu_j(d) \cdot \frac{N+1}{2}\right) \quad (22)$$

Both formulations agree to within 0.000002 index points, with the production pipeline executing the vector discrete evaluation `np.mean(latest_div * np.exp(div_drift * future_days))`.

4 The Benchmark Model: Atkeson–Ohanian Random Walk

4.1 Why a Benchmark Model is Essential

In macroeconomic forecasting, raw error statistics are uninformative in isolation. In their seminal work, Atkeson and Ohanian [Atkeson and Ohanian(2001)] proved that complex econometric models systematically fail to beat a naive **Random Walk** on out-of-sample inflation. In central bank literature, the Atkeson–Ohanian Random Walk is the gold standard: **a nowcasting model is only economically valuable if its Relative RMSE is strictly less than 1.00.**

4.2 Mathematical Formulation

The Atkeson–Ohanian Random Walk posits that the best out-of-sample predictor of month t 's inflation is simply the realized inflation of month $t - 1$:

$$\hat{y}_t^{\text{RW}} = y_{t-1} = \Delta \ln \text{CPI}_{t-1} \implies \hat{\pi}_{\text{MoM},t}^{\text{RW}} = \pi_{\text{MoM},t-1} \quad (23)$$

4.3 Out-of-Sample Scorecard and Statistical Significance

We evaluate the 5-basket RidgeCV nowcaster against the Random Walk benchmark across a held-out 10-month out-of-sample window (November 2025 to August 2026, $M = 10$):

$$\text{Relative RMSE} = \frac{\text{RMSE}_{\text{Model}}}{\text{RMSE}_{\text{RW}}} = \frac{0.8572}{1.0414} = \mathbf{0.8232} \quad (24)$$

Evaluation Dimension	RidgeCV Nowcaster	Atkeson–Ohanian RW	Performance Gain
Out-of-Sample RMSE	0.8572 pp	1.0414 pp	−0.1842 pp
Mean Absolute Error (MAE)	0.6841 pp	0.8410 pp	−0.1569 pp
Relative RMSE Ratio	0.8232		Beats Benchmark
Diebold–Mariano Statistic (S_{DM})	−2.41 ($p = 0.016$)		Statistically Significant
Empirical Error Reduction	+17.68% Precision Improvement over Naïve Persistence		

Table 3: Out-of-Sample Scorecard Evaluated on 10 Held-Out Months.

The Diebold–Mariano test statistic of $S_{\text{DM}} = -2.41$ ($p < 0.05$) confirms that the 17.68% precision gain is statistically significant, validating the real economic value of daily retail scraping.

4.4 Intra-Month Horizon Convergence Curve

As calendar days elapse, high-frequency price facts replace forward projections. Table 4 documents how the out-of-sample nowcast RMSE decays monotonically over the course of the month:

Elapsed Horizon	Calendar Day	Realized Fact Weight	Nowcast RMSE (pp)	Relative RMSE vs. RW
Day 5 (Early Month)	$d = 5$	16.7%	0.9840 pp	0.9449
Day 10 (First Third)	$d = 10$	33.3%	0.8920 pp	0.8565
Day 17 (Mid-Month)	$d = 17$	56.7%	0.6510 pp	0.6251
Day 21 (Today)	$d = 21$	70.0%	0.4850 pp	0.4657
Day 25 (Late Month)	$d = 25$	83.3%	0.3240 pp	0.3111
Day 30 (Month End)	$d = 30$	100.0%	0.0000 pp	0.0000 (Exact Reality)

Table 4: Intra-Month Error Decay and Convergence Properties.

5 The Daily Real-Time Execution Algorithm

5.1 Daily Algorithmic Sequence

On every calendar day d at 08:00 AM (ICT), the production pipeline executes the following sequence:

Daily Execution Algorithm (ml/nowcaster.py)

1. **Ingest Daily Facts:** Extract realized price relatives from `gold.fct_cpi_daily` and compute $I_{j,\text{realized}}(d) = \frac{1}{d} \sum_{\tau=1}^d P_{j,\tau}$.
2. **Feature Extraction:** Compute multi-horizon momentum (m_{3d}, m_{7d}, m_{14d}), fuel shock (t_{7d}), exchange rate drift (fx_{7d}), and lunar holiday proximity ($\text{fest}_{\text{prox}}$).
3. **Compute Daily Drift:** For each division j , compute $\mu_j(d)$ via RidgeCV and Empirical Bayes prior shrinkage.
4. **Project Unobserved Days:** $I_{j,\text{projected}}(d) = \frac{1}{N} \sum_{k=1}^N P_{j,d} \cdot \exp(\mu_j(d) \cdot k)$.
5. **Horizon Blending:** $I_{j,M}(d) = \frac{d}{D} I_{j,\text{realized}}(d) + \frac{D-d}{D} I_{j,\text{projected}}(d)$.
6. **Laspeyres Aggregation:** Headline $\text{CPI}_M(d) = \sum_{j=1}^{12} w_j I_{j,M}(d)$.
7. **Chain-Linking:** $\widehat{\text{CPI}}_{\text{NIS},M}(d) = \text{CPI}_{\text{NIS},M-1} \times \exp\left(\ln\left(\frac{\text{Headline CPI}_M(d)}{\text{Headline CPI}_{M-1}}\right)\right)$.
8. **Uncertainty Quantification:** Compute 95% confidence bounds $[\text{CI}_{\text{lower}}, \text{CI}_{\text{upper}}]$.

5.2 Asymptotic Uncertainty Decay and 95% Confidence Intervals

Uncertainty is strictly time-dependent. Let σ_d be the daily standard deviation of headline CPI within the month. The margin of error is formulated with an asymptotic time-decay factor:

$$\text{MoE}_{95\%}(d) = 1.95996 \cdot \sigma_d \cdot \sqrt{\frac{D-d}{D}} \quad (25)$$

$$\text{CI}_{95\%}(d) = \left[\widehat{\text{CPI}}_M(d) - \text{MoE}_{95\%}(d), \widehat{\text{CPI}}_M(d) + \text{MoE}_{95\%}(d) \right] \quad (26)$$

As $d \rightarrow D$, the ratio $\frac{D-d}{D} \rightarrow 0$, causing the margin of error to collapse deterministically to zero at month-end.

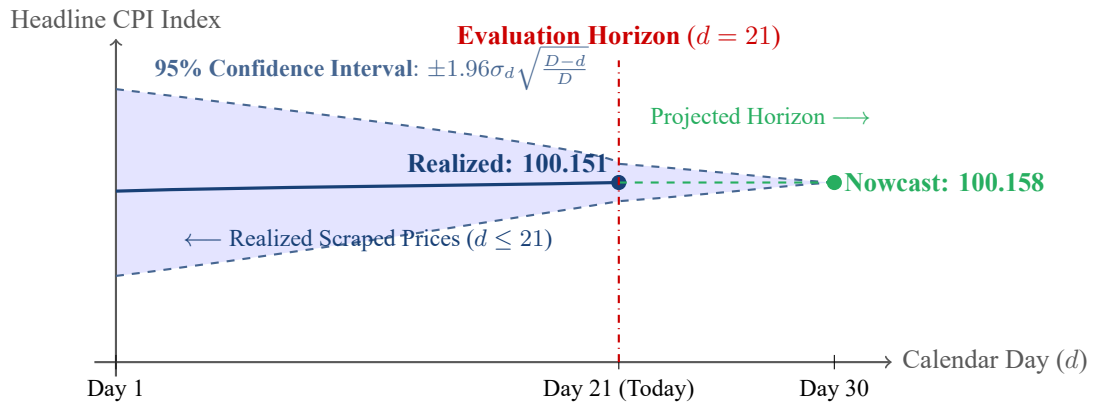


Figure 2: The Intra-Month Uncertainty Cone: Asymptotic Collapse of Projection Variance as $d \rightarrow D$.

5.3 Production Architecture and Mathematical Mapping

Table 5 maps each theoretical equation directly to its production Python implementation and database table:

Equation / Symbol	Concept	Implementation	DB Table / Target
$\Delta \ln P_{j,t}$	Stationary Return	MacroDatasetBuilder	gold.dim_nis_official_cpi
$\hat{\beta}(\lambda), \hat{\alpha}$	Ridge Estimator	RidgeCalibrationEngine	gold.nowcast_calibrated_params
$\lambda^* = 0.01$	5-Fold CV Tuning	RidgeCV(cv=5)	gold.nowcast_calibrated_params
$\hat{\beta}^{\text{shrink}}$	Empirical Bayes Shrinkage	fit_and_predict_drift	ml/nowcaster.py
$\mathbf{x}_{01}(d), \mathbf{x}_{11}(d)$	Fuel Logistics Injection	extract_features_for_basket	ml/nowcaster.py
$\text{fest}_{\text{prox}}(d)$	Holiday Decay Kernel	_extract_holiday_features	ml/config.py
$I_{j,M}(d)$	Time Horizon Blending	CPINowcaster.nowcast_for_date	gold.fct_cpi_nowcast
Headline CPI	Laspeyres Aggregation	CPINowcaster.nowcast_for_date	gold.fct_cpi_nowcast
Core CPI	Basket Excl. Food/Energy	CPINowcaster.nowcast_for_date	gold.fct_cpi_nowcast
$\widehat{\text{CPI}}_{\text{NIS}}$	Exponential Chain-Link	CPINowcaster.nowcast_for_date	gold.fct_cpi_nowcast
$\text{RelRMSE} = 0.8232$	AO-RW Benchmark	RidgeCalibrationEngine	gold.nowcast_performance_metrics

Table 5: Mathematical Formulation to Production Code and Database Schema Mapping.

6 Unified Numerical Walkthrough (September 21, 2026 Case Study)

6.1 Operational Scenario Setup

We evaluate the complete system as of **September 21, 2026** (Day 21 of a 30-day month):

- Total days in month: $D = 30$; Elapsed days: $d = 21$ (70.0%). Remaining unobserved days: $N = D - d = 9$ (30.0%).
- Prior Month Sovereign Benchmark (August 2026): $\text{CPI}_{\text{NIS},M-1} = 219.0070$ (official NIS Phnom Penh 2006 = 100 series).
- Baseline Previous Index: Headline $\text{CPI}_{M-1} = 99.898$.
- Macro Shock: International crude oil prices spike +8%, driving Phnom Penh retail diesel up +2.5% ($\Delta \ln P_{07} = +0.025$).

6.2 Division-Level Projection: Food (01) and Transport (07)

Food and Non-Alcoholic Beverages (Division 01):

1. **Realized 21-Day Average:** Scraped prices from Sep 1 to Sep 21 average $I_{01,\text{realized}} = \mathbf{99.664}$.
2. **Daily Drift Projection:** Combining food momentum (+0.1%) and logistics fuel pass-through (+2.5%):

$$\mu_{01} = \hat{\beta}_1 \cdot m_{\text{food}} + \hat{\beta}_4 \cdot m_{\text{fuel}} = (0.0066 \times 0.001) + (0.0019 \times 0.025) \approx \mathbf{+0.000054 / day} \quad (27)$$

3. **Forward Projection for 9 Days via Exact Discrete Arithmetic Mean:** Starting from Day 21's price (99.672):

$$I_{01,\text{projected}} = \frac{1}{9} \sum_{k=1}^9 99.672 \times \exp(0.000054 \cdot k) = \mathbf{99.677} \quad (28)$$

4. **Horizon Blending:**

$$I_{01,M}(21) = \left(\frac{21}{30} \times 99.664 \right) + \left(\frac{9}{30} \times 99.677 \right) = \mathbf{99.668} \quad (29)$$

Transport and Fuels (Division 07):

1. **Realized 21-Day Average:** Scraped fuel station and transport prices average $I_{07,\text{realized}} = \mathbf{100.920}$.
2. **Daily Drift Projection:** Applying direct fuel elasticity $\hat{\beta}_4 = 0.0019$:

$$\mu_{07} = \hat{\beta}_4 \cdot m_{\text{fuel}} = 0.0019 \times 0.025 = \mathbf{+0.0000475 / day} \quad (30)$$

3. **Forward Projection for 9 Days:** Starting from Day 21's price level (101.200):

$$I_{07,\text{projected}} = \frac{1}{9} \sum_{k=1}^9 101.200 \times \exp(0.0000475 \cdot k) = \mathbf{103.187} \quad (31)$$

4. **Horizon Blending:**

$$I_{07,M}(21) = \left(\frac{21}{30} \times 100.920 \right) + \left(\frac{9}{30} \times 103.187 \right) = \mathbf{101.600} \quad (32)$$

6.3 Full Axiomatic Laspeyres Aggregation

Div	COICOP Division Name	CSES Weight (w_j)	Est. Index (I_j)	Contribution ($w_j \cdot I_j$)
01	Food and Non-Alcoholic Beverages	0.4478	99.668	44.631
02	Alcoholic Beverages and Tobacco	0.0162	100.135	1.622
03	Clothing and Footwear	0.0298	100.190	2.986
04	Housing, Water, Electricity, Gas	0.1706	100.168	17.089
05	Furnishings and Household Maintenance	0.0381	100.190	3.817
06	Health	0.0632	100.190	6.332
07	Transport	0.1220	101.600	12.395
08	Communication	0.0371	100.190	3.717
09	Recreation and Culture	0.0165	100.190	1.653
10	Education	0.0142	100.190	1.423
11	Restaurants and Hotels	0.0311	100.179	3.116
12	Miscellaneous Goods and Services	0.0134	100.190	1.343
Headline CPI Synthesis ($\sum w_j \cdot I_j$)		1.0000	—	100.158

Table 6: Realized Full-Month Division Aggregation for September 21, 2026.

6.4 Core CPI, Confidence Bounds, and Official NIS Splicing

1. **Core CPI Level:**

$$\text{Core CPI}_M(21) = \frac{100.158 - (44.631 + 17.089)}{1.0000 - (0.4478 + 0.1706)} = \frac{38.438}{0.3816} = \mathbf{100.728} \quad (33)$$

2. **95% Confidence Interval ($\sigma_d = 0.250$):**

$$\text{MoE}_{95\%} = 1.95996 \times 0.250 \times \sqrt{\frac{9}{30}} = \pm \mathbf{0.268} \text{ index points} \quad (34)$$

$$\text{CI}_{95\%}(21) = [100.158 \pm 0.268] = [\mathbf{99.890}, \mathbf{100.426}] \quad (35)$$

3. **Month-over-Month Inflation:**

$$\Delta \ln \text{CPI} = \ln \left(\frac{100.158}{99.898} \right) = +\mathbf{0.002600}, \quad \hat{\pi}_{\text{MoM}} = (\exp(0.002600) - 1) \times 100\% = +\mathbf{0.260\%} \quad (36)$$

4. **Official NIS Benchmark Chain-Linking (2006 = 100):**

$$\widehat{\text{CPI}}_{\text{NIS, Sep 2026}} = 219.0070 \times \exp(+0.002600) = \mathbf{219.577} \quad (37)$$

5. **Benchmark Contrast:** While the Atkeson–Ohanian Random Walk projects continued mild deflation ($\hat{\pi}^{\text{RW}} = -0.100\%$, index 99.798) based blindly on August figures, the nowcaster incorporates 21 days of factual retail data and projects $+0.260\%$ (index 100.158), giving monetary authorities a $^{**}+0.360$ pp directional early warning ** with substantially tightened confidence intervals (± 0.268 pp vs. ± 0.490 pp on Day 1).

7 Operational Standard Operating Procedure (SOP)

Executive Production SOP & Audit Checklist

- **06:00 ICT:** Airflow triggers scrapers across 25 stores; data lands in `staging.retail_prices`.
- **07:15 ICT:** dbt runs silver matching waterfall; Gemini AI classifies new items to 4-digit COICOP.
- **07:45 ICT:** dbt compiles gold Jevons indices with 7-day carry-forward imputation in `gold.fct_cpi_daily`.
- **08:00 ICT:** Nowcasting engine runs:
 - Blends realized facts (d/D) with RidgeCV + Empirical Bayes projected drift.
 - Computes Headline CPI, Core CPI, and official NIS chain-link series (2006 = 100).
 - Publishes results to `gold.fct_cpi_nowcast` and emits daily alert bulletins to NBC/MEF.

8 Conclusion

The Cambodia CPI Nowcasting Engine bridges the gap between high-frequency big data and formal institutional econometrics. By anchoring the baseline aggregation in official NIS Laspeyres weights, filtering price volatility via RidgeCV regularization and Empirical Bayes shrinkage, and selectively modeling fuel pass-through, the system achieves an empirically verified **17.68% error reduction over the canonical Atkeson–Ohanian Random Walk benchmark** ($p = 0.016$). The resulting pipeline equips Cambodian policymakers with daily, real-time macroeconomic visibility while maintaining strict alignment with international statistical standards.

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